

Predictive maintenance of Different devices:-

1. Rotatory Devices:-

1. Machine Type:

Dropdown options — used only to adjust thresholds accordingly. For now, we assume **general rotating machinery** (motors, pumps, compressors) in the 15–300 kW range (Class II as per ISO 10816).

2. Operating Speed (RPM)

- Used for context (not typically thresholded unless it's drastically off from rated RPM).
- You can use it to:
 - Detect **belt slip** (lower RPM than expected)
 - Detect **over-speeding** (can reduce lifespan)

Recommended approach: Compare entered RPM with **rated RPM** (user should provide both).

Condition Deviation from Rated RPM

Good $\pm 2\%$

Warning $\pm 2-5\%$

Critical $> 5\%$

3. Vibration Measurement (mm/s RMS)

Based on ISO 10816 (Class II – medium-size machines, motors 15–300kW)

Condition Vibration Level (mm/s RMS)

Good ≤ 1.12

Satisfactory $1.13 - 2.80$

Unsatisfactory $2.81 - 7.10$

Unacceptable > 7.10

4. Temperature (°C)

Based on typical bearing and motor insulation standards (e.g., Class B/C insulation)

Condition Surface Temperature (°C)

Good $\leq 70^{\circ}\text{C}$

Warning $71\text{--}90^{\circ}\text{C}$

Critical $> 90^{\circ}\text{C}$

◆ For motors: insulation class defines max allowed temp (Class B = 130°C , but safe operating temp is much lower) ◆ For bearings: failure likely above 90°C

5. Noise Level (dB)

Based on machine room environment and acoustic emission studies

Condition Noise Level (dB)

Good $\leq 80\text{ dB}$

Warning $81\text{--}90\text{ dB}$

Critical $> 90\text{ dB}$

Sharp increases or unusual patterns may indicate imbalance, bearing faults, or cavitation.

6. Lubricant Condition

Based on oil analysis best practices

Input Value Condition

Good Normal operation

Degraded Suggest Oil Change

Contaminated Immediate Maintenance Required

You could also add: Water Contamination, Metal Particles, etc. if you allow advanced inputs.

OVERALL REPORT:-

Each parameter can be scored (e.g., 0–10 points), and total health score is computed (e.g., out of 50):

Total Score Condition

45–50 Excellent

35–44 Needs Check

Total Score Condition

< 35 Maintenance Required

Links for the reference of above data taken:-

1. [ISO 10816-1 Standards Chart](#)
2. [Predictive Maintenance by Electrical Signature Analysis](#)
3. [Prediction of Induction Motor Faults Using Machine Learning](#)
4. [Fault Analysis and Predictive Maintenance of Induction Motor Using Machine Learning](#)
5. [Predictive Maintenance of Induction Motors Using Ultra-Low Power Wireless Sensors](#)

2. Industrial Boilers

1. Pressure (Operating Pressure)

Boiler Type	Range	Condition	Threshold (psi)	Notes
Steam Boiler (Low Pressure)	Standard ≤ 15 psi	Good	≤ 15	OSHA low-pressure limit
		Warning	16–25	
		Critical	> 25	Risk of overpressure
Steam Boiler (High Pressure)	Rated 150–350 psi	Good	$\pm 5\%$ of rated	Follow rated value
		Warning	± 5 –10%	
		Critical	$> 10\%$ deviation	Possible PRV fault or scaling
Hot Water Boiler	Up to 160 psi	Good	≤ 160	ASME standard
		Warning	161–180	
		Critical	> 180	

2. Temperature (Water & Flue Gas)

Parameter	Condition	Threshold	Notes
Hot Water Temp ($^{\circ}\text{C}$)	Good	$\leq 121^{\circ}\text{C}$ (250 $^{\circ}\text{F}$)	ASME max
	Warning	122–130 $^{\circ}\text{C}$	

Parameter	Condition	Threshold	Notes
Flue Gas Temp (°C)	Critical	>130°C	Overheating
	Good	Steam temp + 20–50°C	Efficient combustion
	Warning	+51–100°C	Fouling likely
	Critical	>100°C above steam temp	Indicates major heat loss or soot

3. Water Level Monitoring

Condition Description

Good Within normal range

Warning Approaching low/high cutoff

Critical Hits low-water cutoff or dry-fire risk

- ◆ Add auto-checks if boiler has electronic level sensors/logs
- ◆ Manually input abnormal levels for inspection alerts

4. Flue Gas Composition

Gas	Condition	Threshold	Notes
Oxygen (O ₂)	Good	3–5%	Optimal combustion
	Warning	6–8%	Excess air
	Critical	>8%	Severe energy loss
Carbon Monoxide (CO)	Good	≤ 100 ppm	Complete combustion
	Warning	101–400 ppm	Burner tuning issue
	Critical	>400 ppm	Risk of carbon monoxide exposure

5. Vibration

Condition	Threshold (mm/s RMS)	Notes
Good	≤ 1.12	ISO 10816 (Class II)
Satisfactory	1.13 – 2.80	Regular checks recommended
Unsatisfactory	2.81 – 7.10	Bearings, mounts inspection

Condition	Threshold (mm/s RMS)	Notes
Unacceptable	> 7.10	Immediate shutdown likely

6. Fuel Consumption Monitoring

Condition	Description
Good	Stable and within expected trends
Warning	Gradual increase (5–10%) in consumption
Critical	Sudden or >10% increase

- ◆ Benchmark against production load & time
- ◆ Look for burner efficiency degradation or leaks

7. Feedwater Quality

Parameter	Condition	Threshold	Not
TDS (ppm)	Good	< 3000	Prevents scale
	Warning	3000–5000	Needs blowdown
	Critical	> 5000	Risk of scaling/failure
pH	Good	8.5 – 10.5 (Alkaline)	For carbon steel
	Critical	< 7 or > 11	Acidic or too alkaline
Conductivity (µS/cm)	Good	< 2500 (Make-up) < 6000 (Boiler)	Check chemical dosing

Overall report:-

Overall Health Scoring (Optional)

Total Score	Condition
45–50	Excellent
35–44	Needs Check
< 35	Maintenance Needed

Parameter	Condition Threshold	Not
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References:-

1. [Boiler Room Guide by Cleaver-Brooks](#)
2. [EPA Boiler Tune-Up Guide](#)
3. [Importance of Safety Compliance for Boiler Operators](#)
4. [Fundamentals of Industrial Boilers and Steam Generation Systems](#)